

Phase Topology Microscopy of the Hydrogen Atom

A Comprehensive White Paper on Direct Topological Imaging of the Simplest Bound Atomic System

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Based on: Phase Theory and Phase Topology Microscopy manuscripts

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Status note: This white paper is written as a theory-development and instrument-concept document grounded in the uploaded Phase Theory and PTM manuscripts. Phase Theory and PTM are presented here as a proposed framework and proposed modality; the claims require experimental validation before they should be treated as established physics.

Executive Summary

This white paper develops a hydrogen-atom-specific application of Phase Topology Microscopy (PTM), the proposed non-particle imaging modality introduced in the PTM manuscript and grounded in Phase Theory. The hydrogen atom is the decisive first target because it is the simplest stable bound system containing both a lepton-sector defect and a baryonic composite defect. In ordinary quantum mechanics, hydrogen is represented by a proton Coulomb center, an electron wavefunction, and a spectrum of bound states. In the Phase Theory interpretation, the same system becomes a coupled topological object in the universal phase substrate Φ : an electron defect with invariant profile $\Xi_e = (k, QH, t, \chi, R)$ and a proton composite with baryonic knot charge, internal color holonomy neutralization, and a surrounding phase-strain envelope.

PTM is not framed here as a higher-resolution electron microscope. It is framed as a qualitatively different measurement architecture. Instead of firing one defect at another and inferring structure from scattering debris, the Φ -scope launches a coherent reference phase front Φ_{ref} , scans reference winding k_{ref} , measures the local increase in the phase-inconsistency functional $I[\Phi]$, and reconstructs a multi-channel topological portrait of the sample. Applied to atomic hydrogen, the intended output is a resolved topological data cube containing signed winding density, lepton/baryon discrimination, spin-framing texture, generation index, and holonomy structure.

The hydrogen atom is also a disciplined test case because standard physics already predicts its energy levels, charge neutrality, hyperfine splitting, and spatial probability distribution with extraordinary precision. Therefore, PTM must not merely produce an image; it must reproduce known hydrogen phenomenology in the appropriate effective regime while making new topological observables available. The central proposed demonstration is not “seeing an electron orbit.” It is reconstructing the coupled phase topology of the electron-proton bound state while preserving the known nonclassical nature of the electron state.

- **Primary objective.** Define the theoretical object imaged by PTM when the target is hydrogen: not a point electron orbiting a point proton, but a bound phase-topological configuration with distinct lepton, baryon, spin, and holonomy channels.
- **Primary instrument claim.** Specify how a Φ -scope should scan hydrogen using k_{ref} -resolved phase interrogation and reconstruct $\Delta\Phi_{\text{signal}}(r)$ into a five-channel Ξ data cube.
- **Primary validation claim.** Show how the PTM hydrogen program can be validated against ordinary spectroscopy, charge neutrality, hyperfine structure, Stark/Zee-man perturbations, isotope shifts, and eventual single-defect imaging.
- **Primary falsification claim.** State conditions under which the PTM interpretation of hydrogen would fail: no reproducible topological contrast, no signed winding channel, no distinction between lepton and baryon sectors, or reconstructions that contradict the known hydrogen spectrum without a compensating, testable correction.

1. Purpose and Scope

The purpose of this paper is to translate the general PTM proposal into a concrete hydrogen-atom white paper. Hydrogen is not chosen because it is large or easy to image in the conventional sense. It is chosen because it is theoretically minimal. If PTM cannot provide a coherent account of hydrogen, it cannot credibly claim to image more complex atoms, molecules, condensed matter, biological macromolecules, or dark-sector defects.

The scope is threefold. First, the paper states the theoretical representation of hydrogen in Phase Theory notation. Second, it specifies the measurement architecture by which a Φ -scope would probe hydrogen without defect-defect collision. Third, it defines a staged validation and falsification program that uses conventional hydrogen observables as calibration anchors while seeking new topological observables unavailable to optical, electron, X-ray, or scanning-probe methods.

The paper deliberately separates three levels of claim: established hydrogen physics; Phase Theory reinterpretation; and PTM engineering projections. Established hydrogen physics includes the Coulomb binding energy scale, Bohr radius scale, spectral transitions, spin coupling, and charge neutrality. The Phase Theory layer interprets those facts as emergent behavior of a coupled phase-topological configuration. The PTM layer proposes an instrument that could directly measure components of that configuration. Confusing these layers would weaken the proposal; keeping them separate makes the white paper experimentally actionable.

2. Source-Theory Foundation

2.1 Phase Theory premises used in this white paper

Phase Theory begins from the postulate that phase is the sole fundamental constituent of physical reality. Space, time, matter, light, and geometry are treated as emergent structures arising from organization, topology, and consistency of a phase-bearing substrate $\Phi : M \rightarrow T$. The target manifold used in the parent theory is $T = U(1) \times SU(2) \times SU(3)$, aligning the target-space structure with the Standard Model gauge sectors while interpreting those sectors topologically rather than as independent fundamental fields.

The key dynamical object is the phase-inconsistency functional:

$$I[\Phi] = \int_M \rho(\Phi, D\Phi) d\mu$$

with density

$$\rho(\Phi, D\Phi) = 1/2 K_{ab}(\Phi) D\mu\Phi^a D\mu\Phi^b + V(\Phi) + \lambda \text{Tr}(F_{\mu\nu} F^{\mu\nu})$$

In the hydrogen application, this functional is used twice: first as the variational object whose constrained minimizers define stable electron and proton sectors, and second as the imaging object whose local perturbation by a reference phase front produces the PTM signal. The key conceptual step is that mass is not treated as an independent primitive. It is interpreted as irreducible phase-inconsistency energy associated with a stable topological sector.

2.2 PTM premises used in this white paper

The PTM manuscript defines the Φ -scope as a system that launches a coherent reference phase front Φ_{ref} with controlled winding number, couples that reference front to the sample phase configuration Φ_{sample} , and measures the resulting increase in phase inconsistency. In the weak-strain regime, the proposed signal is:

$$\Delta\Phi_{\text{signal}} = \kappa \int \rho I(\mathbf{r}) dV$$

with a pixel recorded when

$$\Delta\Phi_{\text{signal}}(r) > \delta_{\text{min}} = \pi/k_{\text{max}}$$

For hydrogen, the measured quantity $\rho_I(r)$ is not merely electron density and not merely proton density. It is the local inconsistency density of the bound electron-proton configuration evaluated against the chosen reference winding sector. A complete hydrogen PTM experiment therefore requires a multi-pass scan over k_{ref} and additional reconstruction priors to separate electron winding, proton baryonic topology, spin framing, and holonomy components.

3. Why Hydrogen Is the Natural First Atomic Target

The hydrogen atom is the simplest atom, but it is not topologically trivial. It contains a negatively charged lepton-sector defect, a positively charged baryonic composite defect, a neutral bound-state exterior, internal spin degrees of freedom, proton substructure, and a precisely known spectral response. These features make hydrogen simultaneously minimal and demanding.

A successful PTM treatment of hydrogen must satisfy five constraints:

1. It must recover charge neutrality in the far field: the combined electron-proton configuration must have no net long-range $U(1)$ winding outside the atom, although local signed winding structure remains present inside the bound state.
2. It must recover the effective quantum electron distribution: the PTM image cannot imply a classical planetary orbit; it must reconstruct a bound phase envelope consistent with the hydrogen orbital state being prepared.
3. It must distinguish the electron as lepton-sector topology from the proton as baryonic composite topology.
4. It must encode spin-framing structure sufficiently to address hyperfine coupling between electron and proton spin sectors.
5. It must preserve known spectroscopy while offering new topological observables, otherwise it is either redundant with conventional methods or inconsistent with established data.

Hydrogen also provides a clean hierarchy of increasingly difficult targets: free electron, free proton or proton proxy, neutral atomic hydrogen, deuterium, muonic hydrogen, Rydberg hydrogen, hydrogen under electric or magnetic fields, and hydrogen in molecular or condensed environments. This hierarchy can be converted directly into a validation roadmap.

4. Standard Hydrogen Physics as Calibration Anchor

The ordinary hydrogen atom is described, at leading order, by a nonrelativistic electron in a Coulomb potential generated by a proton. The characteristic length is the Bohr radius $a_0 \approx 5.29 \times 10^{-11}$ m. The characteristic binding energy is 13.6 eV for the ground state. Corrections include reduced mass, fine structure, Lamb shift, hyperfine splitting, nuclear size, and recoil. These corrections are essential not because PTM is a spectrometer in the usual sense, but because they provide hard calibration tests for any claimed topological reconstruction.

In conventional quantum mechanics, the electron is represented by a wavefunction $\psi_{nlm}(r, \theta, \phi)$, not by a localized bead orbiting the proton. The probability density $|\psi|^2$ is spherically symmetric for the 1s state, has angular structure for p and higher orbitals, and changes under external perturbations. PTM must be consistent with that structure. A correct PTM hydrogen image should not show a classical

circular orbit in the 1s state; it should show a topological phase-inconsistency envelope whose reconstruction, when reduced to the effective quantum regime, corresponds to the known 1s distribution.

The proton is not elementary in standard physics. It is a color-neutral baryon containing quarks and gluonic structure. In the PTM view, this means hydrogen offers both an elementary lepton-sector signature and a baryonic composite signature in one minimal atom. This duality is the chief reason hydrogen is more powerful as a PTM target than a free electron alone.

5. Phase-Topological Model of the Hydrogen Atom

5.1 Electron sector

In the notation used by the source manuscripts, an elementary particle species is labeled by a topological invariant quintuple:

$$\underline{E} = (k, QH, t, \chi, R)$$

For the electron, the working assignment is:

$$\underline{E_e} = (-1, 0, 1, \chi^{1/2}, \text{trivial})$$

Here $k = -1$ encodes negative U(1) winding, $QH = 0$ indicates lepton-sector topology rather than baryonic knot charge, $t = 1$ indicates first generation, $\chi^{1/2}$ indicates spin-half framing, and R is trivial with respect to color holonomy. In a bound hydrogen atom, the electron sector is not an isolated defect. It is phase-coupled to the proton sector, and its experimentally accessible profile is a bound-state distribution rather than a sharp classical trajectory.

5.2 Proton sector

For the proton, the topological description is more complex. As a color-neutral baryonic composite, the proton should present net positive U(1) winding in the electromagnetic channel, nonzero baryonic knot structure, and internally neutralized SU(3) holonomy at the color-singlet boundary. A schematic proton-level assignment can be written:

$$\underline{E_p} \approx (+1, QH, \text{baryon}, 1, \chi_p, R_{\text{color-neutral exterior}})$$

The approximation sign is important. The proton is not a single elementary topological tube in the same sense as the electron. It is a composite knot/holonomy object whose exterior may be color neutral while its interior contains nontrivial holonomy structure. PTM applied to hydrogen should therefore ask two distinct questions: what is the exterior atomic topological signature, and can the instrument, in principle, resolve any proton-internal holonomy contribution? First-generation hydrogen PTM likely targets the former; later PTM generations may attempt the latter.

5.3 Bound-state topology

Hydrogen is not merely the sum $\underline{E_e} + \underline{E_p}$. It is a constrained joint minimizer of the phase-inconsistency functional within a combined topological sector:

$$\Phi H = \arg \min_{\Phi \in \sigma(e,p,bound)} I[\Phi]$$

subject to charge neutrality, finite coherence, spin-coupling constraints, and the bound-state preparation. The electron and proton sectors produce a neutral exterior in the far field while maintaining nontrivial local topology in the interior. The observable hydrogen atom is therefore a nested structure:

- an outer neutral phase-strain envelope corresponding to the atom as a whole;
- an electron-sector winding distribution whose effective projection reproduces the prepared orbital state;
- a baryonic core signature associated with the proton;
- spin-framing correlations producing hyperfine structure;
- a possible weak holonomy residue associated with the proton's composite internal topology, screened at the atomic exterior.

Hydrogen component	Conventional description	PTM / Phase Theory description	Primary PTM channel
Electron	Lepton with charge -e and spin 1/2; bound-state wavefunction	Negative U(1) winding defect with QH = 0, t = 1, spin-half framing	k-map, χ -map, t-map
Proton	Baryon with charge +e and spin 1/2; composite QCD structure	Composite baryonic topological core with net positive winding and color-neutral exterior holonomy	k-map, QH-map, R-map, χ -map
Atomic bond	Coulombic bound state with quantized spectrum	Joint phase-consistency minimum of electron and proton sectors	ΔI map and reconstructed bound envelope
Hyperfine structure	Electron-proton spin coupling	Correlation between electron χ framing and proton χ /baryonic framing	χ -map correlation analysis
Charge neutrality	Vanishing net exterior charge	Cancellation of exterior U(1) winding while retaining local signed winding	k-map integral over far-field boundary

6. PTM Observable for Hydrogen

The hydrogen PTM observable is the phase shift induced when Φ_{ref} is brought into controlled overlap with the hydrogen sample configuration. For a prepared hydrogen state H, write:

$$\Phi_{\text{sample}} = \Phi H = \Phi p \oplus \Phi e \oplus \Phi_{\text{bind}}$$

where Φ_p is the baryonic core contribution, Φ_e is the electron-sector contribution, and Φ_{bind} is the bound-state coherence and strain contribution that cannot be reduced to either constituent alone. The measured weak-strain signal is:

$$\Delta\Phi H(k_{\text{ref}}, r) = \kappa \int V(r) \rho I[\Phi_{\text{ref}}(k_{\text{ref}}), \Phi H] dV$$

The dependence on k_{ref} is central. A single k_{ref} image cannot reconstruct hydrogen. Instead, a scan over reference winding produces a response spectrum at each pixel or voxel. The inversion task is to find the most probable topological field $\Xi H(r)$ compatible with the measured $\Delta\Phi H(k_{\text{ref}}, r)$ across the scan sequence.

6.1 Expected channel signatures

- **Signed winding channel.** The electron-sector contribution should appear with negative polarity relative to positive proton-sector winding. Because the atom is neutral, the integrated exterior k -map should approach zero at sufficiently large boundary radius.
- **Lepton/baryon channel.** The electron contribution should reconstruct with $QH = 0$, while the proton core should reconstruct with baryonic knot charge or baryon-associated topological complexity.
- **Generation channel.** Ordinary hydrogen should be dominated by $t = 1$. Muonic hydrogen or exotic excited targets provide later tests of t -channel discrimination.
- **Spin-framing channel.** Prepared hyperfine states should alter the χ -correlation map without changing the gross k -map.
- **Holonomy channel.** The exterior atom should be color-neutral, while advanced reconstructions may search for proton-core holonomy signatures screened at larger radius.

6.2 What PTM should not show

A scientifically credible hydrogen PTM image must avoid a common misconception: it should not show the electron as a tiny planet orbiting the proton. A 1s hydrogen atom has no classical orbital path. If PTM is consistent with quantum mechanics as an emergent effective regime, the 1s reconstruction should present a spherically symmetric bound phase envelope in the electron-sector response, not a ring trajectory. For p and higher prepared states, angular nodes and phase structure should appear in the reconstructed topology, but again as state-dependent phase structure rather than a literal orbit.

7. Measurement Architecture for Atomic Hydrogen

7.1 Target preparation

Atomic hydrogen is difficult to immobilize without perturbation. The first hydrogen-specific PTM experiments should therefore proceed through staged targets rather than beginning with a free room-temperature atom. Candidate target classes include magnetically trapped ultracold atomic hydrogen, hydrogen adsorbed on cryogenic weakly interacting substrates, ion-trap-adjacent proton/electron calibration systems, Rydberg hydrogen ensembles, and muonic hydrogen only in later specialized programs. The guiding principle is to preserve the relevant phase topology while controlling motion and environmental decoherence.

7.2 Reference phase scan

The Φ -scope should perform a multi-pass scan over k_{ref} . Negative, zero, and positive reference winding values are required to separate electron-sector and proton-sector contributions. The scan should include low-amplitude non-destructive passes to determine baseline drift, followed by k_{ref} -resolved spectroscopy at fixed dwell times. Each pass produces a phase-shift map $\Delta\Phi(k_{\text{ref}}, r)$. The complete data cube is indexed by spatial coordinate, reference winding, detector quadrature, and preparation state.

7.3 Detector and reconstruction requirements

The source manuscripts identify heterodyne atom interferometry and SQUID gradiometer arrays as detector options. For hydrogen, atom interferometry is attractive for phase sensitivity, while SQUID-like detection may be useful in calibration against superconducting vortex targets before moving to atoms. The hydrogen target imposes additional requirements: minimal electromagnetic perturbation, ultrahigh vacuum compatibility, cryogenic or ultracold operating modes, and synchronized state preparation for hyperfine or Rydberg tests.

Reconstruction should be Bayesian or variational. The forward model maps candidate $\Xi H(r)$ fields into predicted $\Delta\Phi H(k_{\text{ref}}, r)$. The inverse model should impose priors from hydrogen physics: charge neutrality, spherical symmetry for 1s, known angular structure for prepared p states, hyperfine state labels, and finite support of proton core effects. The reconstruction cost function should combine data fidelity, topological discreteness, smoothness of coherent regions, and penalties for unphysical net charge.

Subsystem	Hydrogen-specific requirement	Risk	Mitigation
Phase source	Stable k_{ref} scan with phase drift below the operational δ_{min} budget	Reference winding drift can mimic charge contrast	Interleaved zero-sample calibration and active phase lock
Coherence lens	Focus without ionizing or mechanically disturbing trapped hydrogen	Lens fields may perturb atomic state	Use low-field phase lens profiles; compare with field-free spectroscopy controls
Sample stage	Hold hydrogen in a known quantum state	Substrate or trap may alter topology	Use multiple target preparations and require reproducible invariant signatures
Detector	Resolve $\Delta\Phi$ below pixel threshold with low back-action	Detector noise can masquerade as subthreshold topology	Use null targets, deuterium comparison, and repeated k_{ref} reversals
Reconstructor	Separate electron, proton, binding, and spin contributions	Inverse problem underdetermined	Use multi- k_{ref} scans, state-preparation priors, and blinded synthetic-data tests

8. Data Products

The primary data product is not a conventional grayscale micrograph. It is a five-channel topological data cube with uncertainty estimates:

- $kH(r)$: signed winding density, including local electron/proton polarity and far-field neutrality check.
- $QH, H(r)$: knot-charge or baryon/lepton discriminator, expected to isolate the proton core from the electron sector.
- $tH(r)$: generation/excitation map, expected to remain $t = 1$ for ordinary hydrogen.
- $\chi H(r)$: spin-framing texture and electron-proton spin correlation map.
- $RH(r)$: holonomy map, expected to be color-neutral in the exterior with possible proton-core structure in advanced regimes.

A secondary data product is the reduced effective image: the projection of the topological reconstruction into quantities that can be compared with conventional hydrogen observables. Examples include radial electron-sector envelope, spherical symmetry metrics, hyperfine-state contrast, Stark/Zee-man response, and isotope shifts. This reduced image is crucial because it allows PTM to be tested against established physics before its unique topological channels are accepted.

8.1 Proposed hydrogen PTM image layers

Layer	Description	Acceptance criterion for 1s hydrogen
Topological charge layer	Signed k-map after multi-kref inversion	Net integrated exterior charge ≈ 0 ; local electron/proton polarity resolved or statistically inferred
Electron envelope layer	Projection of electron-sector phase topology	Spherical symmetry within uncertainty; characteristic length scale consistent with a_0 after calibration
Baryonic core layer	QH and R-channel signal associated with proton	Localized baryonic signature at atomic center; absent in electron-only control
Spin correlation layer	χ -map dependence on hyperfine preparation	Changes between hyperfine states without gross change in charge envelope
Perturbation response layer	Difference maps under electric or magnetic fields	Stark/Zee-man symmetries consistent with standard selection rules in effective projection

9. Validation Program

9.1 Stage 0: Instrument calibration before hydrogen

Before hydrogen is attempted, the Φ -scope should be calibrated against known topological structures. The PTM manuscript proposes Abrikosov vortex lattices in niobium as an early calibration target. This

is appropriate because the vortex lattice supplies known discrete winding. The immediate goal is to demonstrate that the instrument can measure $\delta_{\min}$ -like phase steps and distinguish topological signal from smooth conventional field profiles.

9.2 Stage 1: Single-sector calibration

The first hydrogen-adjacent calibration should isolate a negative electron-sector signal and a positive proton-sector or ion-sector signal separately. The purpose is not yet to reconstruct a neutral atom. It is to verify sign sensitivity and demonstrate that k_{ref} reversal flips the expected channel response. Failure here would undermine the hydrogen program before bound-state complexity is introduced.

9.3 Stage 2: Neutral hydrogen ground state

The first true hydrogen experiment should target the 1s ground state. The acceptance criteria are: stable neutral exterior k -map, localized baryonic core response, electron-sector envelope consistent with spherical symmetry, and reproducibility across repeated preparations. The strongest early positive result would be simultaneous recovery of far-field neutrality and internal signed winding structure.

9.4 Stage 3: Perturbed hydrogen

After the ground-state measurement, controlled perturbations should be introduced. Electric fields test Stark response; magnetic fields test Zeeman and spin-framing response; microwave preparation tests hyperfine-state dependence; optical excitation tests orbital-state reconstructions. These experiments are critical because they convert a static image claim into a dynamical, state-dependent measurement program.

9.5 Stage 4: Isotope and exotic-atom comparisons

Deuterium should alter the baryonic/nuclear core and reduced-mass effective projection while preserving electron-sector charge. Muonic hydrogen should replace the electron-sector lepton defect with a second-generation lepton-sector defect, changing t and the bound-state radius. These comparisons provide high-value discrimination tests for the t -map and baryonic core reconstruction.

Experiment	Primary question	Expected PTM signature	Falsification pressure
Abrikosov vortex calibration	Can PTM resolve discrete winding?	Integer k steps and reproducible $\delta_{\min}$ threshold	High for instrument concept
Electron/proton controls	Can sign and lepton/baryon channels separate?	Opposite k polarity; QH separation	High for hydrogen imaging
1s hydrogen	Can PTM reconstruct neutral bound topology?	Neutral exterior, spherical electron envelope, localized baryonic core	Very high
Hyperfine preparation	Can χ -map detect spin-framing correlations?	State-dependent χ differences with stable k -map	Medium-high
Stark/Zee-man tests	Does reconstruction transform correctly under fields?	Symmetry-consistent deformation of envelope and spin	Very high

		texture	
Deuterium/muonic hydrogen	Can nuclear/excitation channels discriminate?	Core and t-channel shifts matching target identity	High

10. Comparison with Existing Methods

Conventional methods already probe hydrogen extremely well, but they do so indirectly. Optical spectroscopy measures transition energies. Microwave spectroscopy measures hyperfine splitting. Scattering experiments infer charge and form-factor information. Scanning probes can image orbitals in atoms and molecules on surfaces, but only in environment-dependent, effective-electronic-density terms. PTM claims a different observable: the direct topological decomposition of the bound-state configuration.

The value proposition is therefore not simply smaller pixels. The value proposition is new channel access. If PTM can distinguish k , QH , t , χ , and R contributions in a hydrogen target, it would provide a way to image the topological identity of atomic constituents rather than only their effective charge density or spectral transitions. The hydrogen atom becomes the bridge between conventional quantum observables and the proposed topological ontology.

Method	What it measures in hydrogen	Limitation relative to PTM claim
Optical spectroscopy	Energy differences between states	No direct spatial topological reconstruction
Microwave hyperfine spectroscopy	Spin-coupled transition frequency	No image of spin-framing texture
Electron scattering	Charge/form-factor information	Uses defect-defect collision and can disturb target
STM/AFM on surfaces	Effective local density or force response	Requires surface environment and does not access full Ξ
PTM	Proposed phase-inconsistency/topological channels	Unvalidated; requires difficult coherence engineering and inverse reconstruction

11. Risk, Limitations, and Falsifiability

The hydrogen PTM program has major theoretical and engineering risks. The theoretical risk is that Phase Theory may not be correct or may not couple to laboratory phase references as proposed. The engineering risk is that the coherence budget needed for atomic PTM may exceed practical limits. The reconstruction risk is that multiple topological configurations may fit the same $\Delta\Phi$ data unless the experiment includes sufficient kref diversity, state preparation, and external validation.

11.1 Hard falsification criteria

- No reproducible $\Delta\Phi$ response above calibrated noise for known topological calibration targets.
- No sign reversal or signed contrast under kref reversal in single-sector charge controls.

- No distinguishable response between lepton-sector and baryonic-sector controls after accounting for mass, charge, and ordinary electromagnetic effects.
- Hydrogen reconstructions that imply persistent net exterior charge for a neutral atom.
- 1s hydrogen reconstructions that show classical orbital paths rather than a quantum-compatible envelope.
- Hyperfine, Stark, or Zeeman preparation changes that are absent in PTM despite adequate sensitivity.
- A t-map that cannot distinguish ordinary hydrogen from muonic hydrogen under controlled conditions.

11.2 Non-falsifying nulls

Some null results would constrain but not immediately falsify the broader theory. For example, failure to resolve proton-internal holonomy in first-generation PTM may simply reflect insufficient resolution or reconstruction bandwidth. Failure to image free hydrogen without perturbation may reflect target-preparation limitations. The strongest falsification requires failure on calibration targets, sign-channel tests, or contradictions with established hydrogen symmetries in a regime where the instrument demonstrably has sufficient sensitivity.

12. Development Roadmap for Hydrogen PTM

Phase	Milestone	Hydrogen-specific deliverable
H0: Theory and simulation	Forward model for ΦH and synthetic Ξ data cubes	Open benchmark dataset for 1s, 2p, hyperfine, Stark, and Zeeman cases
H1: Calibration hardware	Vortex and charge-sector calibration	Validated k-channel response and $\delta_{\min}$ threshold
H2: Single-defect controls	Electron/proton proxy measurements	Signed winding and lepton/baryon discrimination controls
H3: Neutral hydrogen	Ground-state atomic hydrogen imaging	Neutral exterior + internal signed/channel reconstruction
H4: State-dependent hydrogen	Excited and perturbed states	Orbital, hyperfine, Stark, and Zeeman difference maps
H5: Isotope/exotic validation	Deuterium and muonic hydrogen comparisons	Baryonic core and t-channel discrimination

The roadmap should be executed with blinded reconstruction whenever possible. Synthetic data generated from known candidate topologies should be mixed with experimental data to prevent overfitting. Each phase should define pre-registered acceptance criteria before data collection begins.

13. Conclusion

Phase Topology Microscopy of the hydrogen atom is the natural first atomic test of the PTM program. Hydrogen is simple enough to model, precise enough to falsify, and rich enough to test every essential PTM channel. A successful program would not replace ordinary hydrogen spectroscopy; it would add a

new class of observable: the spatially reconstructed topological identity of the bound electron-proton system.

The decisive claim is not that PTM can draw a prettier picture of hydrogen. The decisive claim is that hydrogen has a phase-topological structure beneath its effective quantum description, and that a Φ -scope can couple to that structure without relying on particle-defect scattering. If the claim is right, the first topological image of hydrogen would become a foundation stone for a broader program: imaging matter as phase topology rather than as objects in space. If the claim is wrong, the hydrogen program should reveal that quickly through calibration failure, channel nonseparability, or contradiction with known hydrogen observables. Either outcome is scientifically useful.

Appendix A. Hydrogen PTM Notation

Symbol	Meaning
Φ	Universal phase substrate or sample phase configuration
Φ_{ref}	Controlled reference phase front launched by the Φ -scope
Φ_H	Hydrogen sample phase configuration
$I[\Phi]$	Global phase-inconsistency functional
$\rho I(r)$	Local inconsistency density evaluated against Φ_{ref}
$\Delta\Phi_{\text{signal}}$	Measured phase shift produced by local topological mismatch
δ_{min}	Minimum coherent phase interval, π/k_{max}
Ξ	Topological invariant quintuple (k, QH, t, χ, R)
k	U(1) winding number / signed charge proxy
QH	Knot charge / lepton-baryon discriminator
t	Generation or excitation index
χ	Spin framing / chirality texture
R	Holonomy or gauge representation channel

Appendix B. Recommended White-Paper Figures

The following figures are recommended for the next graphical revision of this document:

1. Schematic of hydrogen as a coupled electron-sector winding envelope plus baryonic core topology.
2. Five-channel Ξ data cube representation of the hydrogen atom.
3. k_{ref} scan sequence and reconstruction pipeline from $\Delta\Phi(k_{\text{ref}}, r)$ to $\Xi_H(r)$.
4. Validation ladder from Abrikosov vortex lattice to 1s hydrogen to muonic hydrogen.
5. Comparison between conventional electron-density representation and PTM topological-channel representation.

References and Source Basis

1. Hanks, M. (2026). Phase Theory: A Unified Theory of Matter, Light, and Geometry. Preprint, January 9, 2026.
2. Hanks, M. (2026). Phase Topology Microscopy: A Non-Particle Imaging Modality Grounded in Phase Theory. Preprint, submitted May 15, 2026.
3. Bohr, N. (1913). On the Constitution of Atoms and Molecules. Philosophical Magazine.
4. Schrödinger, E. (1926). Quantisierung als Eigenwertproblem. Annalen der Physik.
5. Bethe, H. A., and Salpeter, E. E. Quantum Mechanics of One- and Two-Electron Atoms.
6. Cohen-Tannoudji, C., Diu, B., and Laloë, F. Quantum Mechanics.
7. Sakurai, J. J., and Napolitano, J. Modern Quantum Mechanics.
8. Jackson, J. D. Classical Electrodynamics.
9. Abrikosov, A. A. (1957). On the Magnetic Properties of Superconductors of the Second Group.
10. Zurek, W. H. Decoherence and the transition from quantum to classical.